

Name: _____

Class: _____



JURONG JUNIOR COLLEGE

JC2 Preliminary Examinations 2018

MATHEMATICS
Higher 2

Paper 2

9758/02

12 September 2018

3 hours

Additional materials: Answer Paper
 Cover Page
 List of Formulae (MF 26)

READ THESE INSTRUCTIONS FIRST

Write your name and civics class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of **6** printed pages.

[Turn over

Section A: Pure Mathematics [40 marks]

- 1** Given that $f(x) = e^{\sin x}$, use the standard series to find the series expansion for $f(x)$ in the form $a + bx + cx^2 + dx^3$, where a, b, c and d are constants to be determined.

Hence show that the first three non-zero terms for the expansion of $\frac{1}{(e^{\sin x})^2}$ in ascending powers of x is $1 - 2x + 2x^2$. [4]

The function $y = g(x)$ satisfies $4\frac{dy}{dx} = (y+1)^2$ and $y=1$ at $x=0$.

- (i)** By further differentiation, find the series expansion for $g(x)$, up to and including the term in x^3 .

Hence show that when x is small,

$$g(x) - f(x) \approx \frac{1}{4}x^3. \quad [5]$$

- (ii)** By using the result in **(i)**, justify whether $f(x)$ is a good approximation to $g(x)$ for values of x close to zero. [1]

- 2** In 2004, 10 000 cases of obesity among Singaporeans aged 18-30 years old were reported. Each year after that, the number of cases reported increased by 7%. If this pattern were to continue, how many obesity cases would be reported in 2018? Leave your answer to the nearest whole number. [3]

John, who was obese, started on a weight-loss programme. The number of calories he burned in the first week of his exercise regime was a . As the intensity of the exercise regime increased, the number of calories John burned each week was increased by d . On the other hand, the number of calories John consumed each week is a geometric sequence such that the numbers of calories he consumed in the first, second and third week equal the numbers of calories he burned through exercising in the seventh, third and first week respectively.

- (i)** Show that $d = \frac{a}{2}$. [2]

- (ii)** John burned 3000 calories in the seventh week through exercising. Find the least number of weeks required for the total number of calories John burned to exceed the total number of calories he consumed by at least 200 000. [5]

- 3 (i) Using the method of differences, find $\sum_{r=1}^n \frac{1}{r(r+1)}$. [3]

Hence find $\sum_{r=1}^n \left[3^{-r} - \frac{1}{r(r+1)} \right]$. [3]

- (ii) Use your result in part (i) to show

$$\sum_{r=3}^{2N} \left[3^{1-r} - \frac{1}{r(r-1)} \right] = -\frac{1}{3} + \frac{1}{2} \left(\frac{1}{N} - \frac{1}{3^{2N-1}} \right).$$
 [3]

Hence find $\sum_{r=3}^{\infty} \left[3^{1-r} - \frac{1}{r(r-1)} \right]$. [1]

- 4 (i) By using the substitution $x = \frac{1}{3} \sin^2 \theta$, where $0 \leq \theta < \frac{\pi}{2}$, find the exact value of

$$\int_0^{\frac{1}{4}} \sqrt{\frac{x}{1-3x}} \, dx.$$
 [5]

The region R is bounded by the curve $y = \sqrt{\frac{x}{1-3x}}$, the line $y = 1$ and the y -axis.

- (ii) Using your answer in (i), find the exact value of the area of R . [2]

- (iii) Find the volume of revolution when R is rotated completely about the y -axis. Give your answer correct to 4 decimal places. [3]

Section B: Probability and Statistics [60 marks]

- 5 There are ten boys and twelve girls in a school table tennis club. A team of seven boys and seven girls will be selected randomly to represent the school in a table tennis friendly match.

- (i) In how different ways can the team be formed? [2]

- (ii) Jason is the youngest boy and Joyce is the youngest girl in the club. What is the probability that the team includes both Jason and Joyce? [2]

- (iii) Joel is the oldest boy in the club. Given that Joel is selected for the team, what is the probability that the team includes Jason or Joyce, but not both? [4]

[Turn over]

- 6 Alice and Betty each throw a fair cubical die simultaneously.

The random variable X is the larger number shown on the two dice or the common number of the dice if the numbers are equal.

(i) Show that $P(X \leq x) = \left(\frac{x}{6}\right)^2$, for $x = 1, 2, \dots, 6$. [2]

(ii) Deduce that $P(X = x) = \frac{2x-1}{36}$, for $x = 1, 2, \dots, 6$. [1]

(iii) Show that $E(X) = \frac{161}{36}$ and $\text{Var}(X) = \frac{2555}{1296}$. [3]

(iv) Forty independent observations of X are taken. Using a suitable approximation, estimate the probability that the mean of these observations is at least 4.5. [3]

- 7 The number of employees, y , who stay back and continue to work in the office t minutes after 5 pm on a particular day in a company is recorded. The results are shown in the table.

t	15	30	45	60	75	90	105
y	30	19	15	13	12	11	10

(i) Draw a scatter diagram for these values, labeling the axes clearly. [1]

(ii) Find, correct to 4 decimal places, the product moment correlation coefficient between

(a) t and y ,

(b) $\sqrt{t}$ and y ,

(c) $\frac{1}{t}$ and y .

Hence, state with a valid reason, which of the above models is the most appropriate model of the relationship between t and y . [4]

(iii) Using the model you chose in part (ii), find the equation for the relationship between t and y . [2]

(iv) Predict, to the nearest whole number, the number of employees who stay back and continue to work in the office at 7 pm on that particular day. Comment on the reliability of your prediction. [2]

- 8** A jar contains 10 blue and 8 red marbles. Five marbles are randomly drawn from the box, one by one and without replacement.

- (i) Explain why it is inappropriate to model the number of blue marbles by a binomial distribution. [1]
- (ii) Find the probability that exactly three marbles are blue. [3]

Another jar contains 20 blue and 12 red marbles. n marbles are randomly drawn from the box, one by one and with replacement. The number of red marbles drawn is denoted by R .

- (iii) Given that the mean of R is 4.5, find n and $P(R > 4)$. [3]
- (iv) Given instead that $P(R = 0 \text{ or } 1) < 0.01$, write down an inequality for n and find the least value of n . [3]

- 9** Durians and melons are sold by weight. The masses, in kg, of durians and melons are modelled as having independent normal distributions with means and standard deviations as shown in the table.

	Mean Mass	Standard Deviation
Durians	2.1	0.25
Melons	0.6	0.16

Durians are sold at \$15 per kg and melons at \$6 per kg.

- (i) Find the probability that the mass of a randomly chosen durian is less than four times the mass of a randomly chosen melon. [3]
- (ii) Two durians and eight melons are randomly selected. Find the probability that the average mass of these ten fruits exceeds 1 kg. [4]
- (iii) Find the probability that the total selling price of a randomly chosen durian and a randomly chosen melon is less than \$40. [4]
- (iv) Without any further calculation, explain why the probability of the event that both a randomly chosen durian has a selling price less than \$35 and a randomly chosen melon has a selling price less than \$5 is less than the answer to part (iii). [1]

[Turn over

- 10** There was a complaint that the average waiting time for a patient to see a doctor in a local polyclinic is longer than 60 minutes. A public relation officer in the polyclinic investigated the waiting times, x minutes, for 70 randomly chosen patients. The data are summarised by

$$\sum (x - 50) = 1071, \quad \sum (x - 50)^2 = 73158.$$

- (i)** Explain whether the public relation officer should use a 1-tail or a 2-tail test. [1]
- (ii)** Explain why the public relation officer is able to carry out a hypothesis test without knowing anything about the population distribution of the waiting times for the patients to see a doctor. [1]
- (iii)** Find unbiased estimates of the population mean and variance. [2]
- (iv)** Test, at the 5% significance level, whether the complaint is valid. [4]

In another test, using the same set of data and also at the 5% significance level, the hypotheses are as follows:

H_0 : the population mean waiting time is equal to k minutes.

H_1 : the population mean waiting time is not equal to k minutes.

- (v)** Given that the null hypothesis is rejected in favour of the alternative hypothesis, find the set of possible values of k . [4]